Highlights

Open-Vocabulary and Reasoning Segmentation with Vision-Language Foundation Models: A Survey

Author Name

- The survey is organized around technical routes rather than a paper-by-paper chronology.
- The core comparison covers CLIP-based OVSS, diffusion-based segmentation, SAM-based prompting, unified decoders, and MLLM-based reasoning segmentation.
- The outline records where each section should discuss content, figures, datasets, metrics, and cross-method limitations.

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Author Name

School or Department, University or Institution, , , ,

Abstract

This survey outline reviews open-vocabulary and reasoning segmentation with vision-language foundation models. The planned manuscript follows the transition from closed-set segmentation to CLIP-based open-vocabulary segmentation, promptable segmentation with SAM, unified vision-language decoding, and multimodal large language model based reasoning segmentation and pixel grounding. The final manuscript will emphasize method taxonomy, datasets, evaluation metrics, advantages, limitations, and future challenges across representative work from 2023 to 2025.

Keywords: Open Vocabulary segmentation, reasoning segmentation, vision-language models, SAM, CLIP, pixel grounding

1. Introduction

Image segmentation is a fundamental dense prediction problem in computer vision. Classical segmentation tasks are commonly formulated under a closed-set assumption, where the label space is predefined during training and remains fixed during evaluation. Semantic segmentation assigns a category label to each pixel, instance segmentation further separates different object instances, and panoptic segmentation unifies stuff and thing categories into a coherent scene-level representation Long et al. (2015); He et al. (2017); Kirillov et al. (2019). Although these formulations have supported substantial progress in scene understanding, the fixed-category assumption limits their applicability in open-world scenarios, where visual systems must handle unseen categories, long-tail concepts, and user-specified object names beyond the training taxonomy.

Open-vocabulary segmentation has therefore emerged as an important step toward more deployable visual understanding. Different from closed-set segmentation, open-vocabulary segmentation allows the target label space to be defined by text at inference time, such as category names or natural-language prompts. This transition is largely driven by vision-language foundation models, especially CLIP, which learns transferable visual representations from large-scale image-text supervision and enables textual concepts to be used as visual classifiers Radford et al. (2021). Based on this image-text alignment, recent methods such as OVSeg and SAN adapt CLIP-style recognition to dense prediction by combining mask proposals, side adapters, or region-level recognition mechanisms **??**. These methods demonstrate that segmentation can move beyond a fixed training vocabulary, while also revealing the tension between open semantic recognition and accurate pixel-level localization.

Meanwhile, segmentation foundation models and unified vision-language models further reshape the technical landscape. The Segment Anything Model (SAM) reframes segmentation as prompt-conditioned mask generation and shows strong zero-shot transfer across image domains and segmentation prompts **?**. Rather than predicting category labels directly, SAM provides high-quality class-agnostic masks, which makes it a strong spatial prior for subsequent open-vocabulary segmentation pipelines. In parallel, diffusion-based and unified-decoder approaches broaden the task formulation. ODISE exploits representations from text-to-image diffusion models for open-vocabulary panoptic segmentation, while X-Decoder introduces a generalized decoding framework that connects pixel-level segmentation with image-level and language-level outputs **??**. These developments suggest that modern segmentation is increasingly shaped by the interaction between semantic alignment, mask generation, and multimodal decoding.

More recently, the focus has further shifted from category-level openness to language-driven reasoning and pixel grounding. In open-vocabulary segmentation, the target is usually specified by an explicit category name or text label. However, in reasoning segmentation, the target object may be implicit and must be inferred from complex instructions, commonsense knowledge, or scene context. LISA formalizes this setting by introducing reasoning segmentation with large language models, where a special segmentation token is used to connect language understanding with mask prediction Lai et al. (2024). GLaMM extends this direction toward grounded multimodal conversation, where generated phrases can be associated with corresponding

Figure plan: draw a two-dimensional map with years on the horizontal axis and method families on the vertical axis.

Figure 1: Planned overview figure: timeline and taxonomy of open-vocabulary and reasoning segmentation from 2023 to 2025. The figure should place OVSeg, SAN, ODISE, SAM, X-Decoder, PnP-OVSS, LISA, GLaMM, ESC-Net, and Trident along both a temporal axis and a method-family axis.

pixel-level masks Rasheed et al. (2024). This shift indicates that segmentation is becoming not only an open-vocabulary recognition problem, but also a grounding interface between visual perception and natural-language reasoning.

This survey reviews open-vocabulary and reasoning segmentation with vision-language foundation models. Instead of organizing the literature as a paper-by-paper chronology, we structure the survey around major technical routes, including CLIP-based open-vocabulary semantic segmentation, diffusion-based segmentation, SAM-based promptable segmentation, unified vision-language decoders, and MLLM-based reasoning segmentation and pixel grounding. We further discuss extensions to video, 3D, and complex multi-context scenes. Across these method families, the survey compares their task assumptions, model inputs, supervision forms, output types, datasets, evaluation metrics, advantages, and limitations. The goal is to clarify how the field has evolved from closed-set dense prediction to open-vocabulary recognition, and further toward reliable language-guided pixel grounding in open-world visual understanding.

2. Background and Objective

2.1. Closed-set segmentation and its limitation

Writing plan: define the classical dense prediction tasks before introducing open-vocabulary methods. The goal is to clarify what changes when segmentation is no longer restricted to a fixed label set.

2.1.1. Semantic segmentation

Explain that semantic segmentation assigns one class label to each pixel. It is suitable for scene parsing but does not distinguish different instances of the same category.

2.1.2. Instance segmentation

Explain that instance segmentation predicts object-level masks and instance identities, but it normally assumes a predefined set of object categories.

2.1.3. Panoptic segmentation

Explain that panoptic segmentation unifies thing and stuff prediction. Use ODISE as the later open-vocabulary panoptic example (Xu et al., 2023a).

2.2. Open-vocabulary segmentation

Writing plan: define open-vocabulary semantic segmentation as dense classification over categories that may not appear in the segmentation training set. Emphasize that text prompts or category names provide the label space at inference time.

2.2.1. Text-defined label spaces

Explain how category names or prompt templates become text embeddings. This should prepare the reader for CLIP-based methods such as OVSeg and SAN (Liang et al., 2023; Xu et al., 2023b).

2.2.2. Zero-shot and cross-dataset transfer

Explain that open-vocabulary performance is usually measured by transferring from one training dataset to unseen datasets such as ADE20K, PASCAL VOC, PASCAL Context, and COCO-Stuff. Note that mIoU remains common but does not fully capture vocabulary ambiguity.

2.3. Reasoning segmentation and pixel grounding

Writing plan: introduce the shift from category names to complex natural-language instructions. The key distinction is that the target object may be implicit and must be inferred from language, commonsense, scene context, or conversation.

2.3.1. Referring segmentation versus reasoning segmentation

Explain that referring segmentation usually localizes an explicitly described object, whereas reasoning segmentation may require implicit constraints such as "the object that the man is holding". LISA should be introduced here as the representative reasoning segmentation method (Lai et al., 2024).

Figure plan: show how the input changes from a fixed class list, to text labels, to referring expressions, to reasoning queries and multimodal dialogue.

Figure 2: Planned background figure: task hierarchy from closed-set segmentation to open-vocabulary segmentation, referring segmentation, reasoning segmentation, and grounded conversation.

2.3.2. Pixel grounding in multimodal conversation

Explain that pixel grounding connects generated words, phrases, or entities to masks. GLaMM should be positioned as the representative grounded conversation method (Rasheed et al., 2024).

3. Method Taxonomy

3.1. CLIP-based open-vocabulary segmentation

Writing plan: explain why CLIP is attractive for open-vocabulary segmentation and why direct CLIP application is insufficient for pixel-level prediction. The central tension is strong image-text alignment versus weak dense localization.

3.1.1. Mask-adapted region classification

Discuss OVSeg as the canonical two-stage route: class-agnostic mask proposal followed by mask-adapted CLIP classification (Liang et al., 2023). Explain the masked-region distribution shift and why mask prompt tuning or adaptation is needed.

3.1.2. Efficient side adaptation

Discuss SAN as an efficient alternative to repeated proposal classification. Emphasize the side adapter, attention bias, and proposal-wise recognition with a mostly frozen CLIP backbone (Xu et al., 2023b).

3.1.3. Training-free VLM feature mining

Discuss PnP-OVSS as a 2024 training-free route that mines localization cues from off-the-shelf VLMs (Luo et al., 2024). Explain the advantage of removing task-specific segmentation training and the limitation of heuristic sensitivity.

Figure plan: redraw three columns for OVSeg, SAN, and PnP-OVSS: proposal-classification, side-adapter recognition, and training-free feature mining.

Figure 3: Planned figure: comparison of CLIP-based OVSS pipelines.

Figure plan: show a text-to-image diffusion UNet feeding a mask decoder and a text-conditioned open-vocabulary classifier.

Figure 4: Planned figure: ODISE-style diffusion feature pipeline.

3.2. Diffusion-based open-vocabulary segmentation

Writing plan: introduce the idea that text-to-image diffusion models learn spatially meaningful intermediate representations, which can be repurposed for discriminative segmentation.

3.2.1. Diffusion features as spatial-semantic representations

Discuss why the internal UNet features of text-to-image diffusion models can contain object- and part-level semantics. Use ODISE as the main example (Xu et al., 2023a).

3.2.2. Combining generative and discriminative components

Explain how ODISE combines frozen diffusion features, a mask generator, and CLIP-based text classification for open-vocabulary panoptic segmentation (Xu et al., 2023a).

3.2.3. Strengths and costs

Summarize the main advantage as broad spatial-semantic knowledge and the main limitation as high training and inference cost.

3.3. Promptable segmentation foundation models

Writing plan: present SAM as the foundation model that changes segmentation from category prediction to prompt-conditioned mask generation (Kirillov et al., 2023).

Figure plan: start from SAM's image encoder, prompt encoder, and mask decoder, then branch to ESC-Net's CLIP-SAM fusion and Trident's training-free CLIP/DINO/SAM composition.

Figure 5: Planned figure: SAM and SAM-derived OVSS routes.

3.3.1. SAM and promptable segmentation

Explain point, box, and mask prompts; the image encoder, prompt encoder, and mask decoder; and the SA-1B data engine (Kirillov et al., 2023). Stress that SAM produces high-quality class-agnostic masks rather than semantic labels.

3.3.2. SAM-enhanced open-vocabulary segmentation

Discuss ESC-Net as a 2025 example that embeds SAM decoder blocks into one-stage OVSS while retaining CLIP-based category semantics (Lee et al., 2025). The section should explain how SAM supplies spatial priors while CLIP supplies text-category alignment.

3.3.3. Training-free VFM composition

Discuss Trident as a 2025 training-free foundation-model composition route, combining complementary signals such as CLIP, DINO, and SAM (Shi et al., 2025). Highlight the trade-off between no task-specific training and multi-model inference complexity.

3.4. Unified vision-language segmentation models

Writing plan: explain why unified decoders are an intermediate stage between specialized segmentation models and MLLM-based reasoning systems.

3.4.1. Pixel, image, and language outputs in one decoder

Discuss X-Decoder as a generalized decoder for segmentation, referring, retrieval, captioning, and VQA transfer (Zou et al., 2023). Explain how pixel-level and language-level outputs are represented in a shared model.

3.4.2. Generic queries and semantic queries

Explain generic queries, text queries, and semantic query interaction. This part should clarify why X-Decoder is broader than CLIP-based mask classification but still different from MLLM reasoning.

Figure plan: redraw an image encoder, text encoder, and shared decoder with outputs for segmentation masks, retrieval embeddings, captions, and VQA responses.

Figure 6: Planned figure: generalized decoding in X-Decoder.

3.4.3. Transition toward generalist vision-language models

Position X-Decoder as a bridge: it unifies dense and language tasks, but it does not provide the same free-form reasoning and grounded conversation interface as LISA or GLaMM (Lai et al., 2024; Rasheed et al., 2024; Zou et al., 2023).

3.5. MLLM-based reasoning segmentation and pixel grounding

Writing plan: describe the shift from text labels to complex instructions. The key question is how a language token or generated phrase becomes a pixel mask.

3.5.1. Reasoning segmentation with segmentation tokens

Discuss LISA's use of a special segmentation token and its hidden embedding to drive a mask decoder (Lai et al., 2024). Explain the role of MLLM understanding, SAM-style segmentation backends, and instruction-formatted training data.

3.5.2. Grounded conversation generation

Discuss GLaMM as a grounded multimodal model that links generated phrase-level entities to pixel masks (Rasheed et al., 2024). Contrast this with single-target reasoning segmentation.

3.5.3. Risks: hallucination, evaluation, and mask precision

Explain that MLLM-based methods can understand richer language but introduce hallucination risk, heavier training and inference, and more complex evaluation. The subsection should separate reasoning correctness from boundary accuracy.

3.6. Extensions to video, 3D, and complex scenes

Writing plan: keep this subsection shorter because the local ten-paper set mainly covers 2D image segmentation. Use it as a forward-looking bridge to

Figure plan: compare LISA's segmentation-token-to-mask path with GLaMM's phrase-level grounded conversation path.

Figure 7: Planned figure: MLLM-to-mask mechanisms.

Figure plan: draw image, video, 3D/multiview, and multi-context settings, with the added challenge under each setting.

Figure 8: Planned figure: extension path from image segmentation to video, 3D, and multi-context grounding.

video reasoning segmentation, 3D or multiview grounding, and multi-context visual grounding.

3.6.1. Video reasoning segmentation

Explain that video adds temporal consistency, object persistence, motion cues, and memory. Add VRS-HQ or GLUS citations after their source files are added to the project bibliography.

3.6.2. 3D and multiview grounding

Explain that 3D and multiview scenes require spatial consistency across views, camera poses, and object instances. This should be treated as an extension of pixel grounding beyond single images.

3.6.3. Complex multi-context scenes

Explain that multi-image or multi-context grounding stresses MLLM reasoning and object disambiguation. Add MC-Bench or related citations after the corresponding source files are included.

4. Datasets and Evaluation Metrics

4.1. Dataset groups

Writing plan: group datasets by task rather than listing them alphabetically. The tables should show how each dataset supports a different part of the survey argument.

Table 1: Core datasets for open-vocabulary and promptable segmentation.

Dataset	Task	Used by	Main property
SA-1B	Promptable segmentation	SAM (Kirillov et al., 2023)	Large-scale class-agnostic mask data.
ADE20K	Scene parsing and OVSS	SAN, ODISE, ESC-Net, Trident (Xu et al., 2023b,a; Lee et al., 2025; Shi et al., 2025)	Rich scene categories for open-vocabulary transfer.
PASCAL VOC	Semantic segmentation	OVSS methods (Liang et al., 2023; Xu et al., 2023b; Luo et al., 2024)	Compact and common zero-shot benchmark.
PASCAL Context	Semantic segmentation	OVSS methods (Liang et al., 2023; Xu et al., 2023b)	Broader dense semantic labels than VOC.
COCO-Stuff	Semantic segmentation	OVSS training and evaluation (Xu et al., 2023b; Liang et al., 2023)	Stuff categories and dense scene labels.
COCO Panoptic	Panoptic segmentation	ODISE, X-Decoder (Xu et al., 2023a; Zou et al., 2023)	Joint thing and stuff annotation.

4.2. Evaluation metrics

Writing plan: separate segmentation quality, open-vocabulary transfer, reasoning correctness, and computational cost. The final text should avoid treating all methods as if they solve the same task.

4.2.1. Dense segmentation metrics

Discuss mIoU for semantic segmentation, PQ for panoptic segmentation, and AP or mask AP for instance-level outputs. Connect mIoU to OVSeg, SAN, PnP-OVSS, ESC-Net, and Trident (Liang et al., 2023; Xu et al., 2023b; Luo et al., 2024; Lee et al., 2025; Shi et al., 2025); connect PQ/AP to ODISE and X-Decoder (Xu et al., 2023a; Zou et al., 2023).

Table 2: Datasets for language-guided, reasoning, grounding, and future video settings.

Dataset	Task	Used by	Main property
RefCOCO / RefCOCO+ / RefCOCOg	Referring segmentation	LISA, GLaMM, X-Decoder (Lai et al., 2024; Rasheed et al., 2024; Zou et al., 2023)	Language-guided object masks.
ReasonSeg	Reasoning segmentation	LISA (Lai et al., 2024)	Complex implicit queries requiring reasoning.
GranD / GranDf	Grounded conversation	GLaMM (Rasheed et al., 2024)	Phrase-level masks for generated responses.
Video reasoning datasets	Video reasoning segmentation	Future video extensions	Temporal masks and consistency requirements.

Figure plan: map each task family to datasets and metrics, so readers can see why mIoU alone is insufficient for reasoning segmentation and grounded conversation.

Figure 9: Planned figure: dataset and metric map.

4.2.2. Language-guided and reasoning metrics

Discuss cIoU, gIoU, and task-specific grounding scores for referring and reasoning segmentation. Explain that these metrics should be interpreted together with language correctness for LISA and GLaMM (Lai et al., 2024; Rasheed et al., 2024).

4.2.3. Efficiency and deployment metrics

Discuss FPS, GFLOPs, parameter count, memory, and multi-model inference cost. This is important for comparing SAN, PnP-OVSS, ESC-Net, and Trident (Xu et al., 2023b; Luo et al., 2024; Lee et al., 2025; Shi et al., 2025).

Table 3: Comparison of segmentation-centered method families.

Method type	Representative papers	Advantage	Shortcoming
CLIP-based OVSS	OVSeg, SAN, PnP-OVSS (Liang et al., 2023; Xu et al., 2023b; Luo et al., 2024)	Strong open-vocabulary semantics and natural zero-shot category transfer.	CLIP is not designed for dense localization; spatial precision and prompt sensitivity remain issues.
Diffusion-based segmentation	ODISE (Xu et al., 2023a)	Diffusion features provide rich spatial-semantic representations and support panoptic output.	Training and inference are costly, and the system is more complex than lightweight CLIP adapters.
SAM and VFM-based methods	SAM, ESC-Net, Trident (Kirillov et al., 2023; Lee et al., 2025; Shi et al., 2025)	High-quality masks and strong generalization from promptable or compositional foundation models.	Native semantic understanding is limited; additional CLIP, VLM, or MLLM components are required.

5. Comparison of Advantages and Shortcomings

5.1. Cross-method comparison

Writing plan: this is the main synthesis section. Do not describe papers one by one. Compare assumptions, model inputs, supervision, output type, strengths, and weaknesses.

5.2. Trade-off axes

Writing plan: summarize the survey using several axes: semantic openness, mask quality, reasoning complexity, training cost, inference cost, and extensibility.

5.2.1. Semantic openness versus spatial precision

Compare CLIP-based methods with SAM-based methods. CLIP supplies open semantics but weak localization, whereas SAM supplies masks but lacks

Table 4: Comparison of generalist, reasoning, and extension-oriented method families.

Method type	Representative papers	Advantage	Shortcoming
Unified decoder models	X-Decoder (Zou et al., 2023)	Pixel, image, and language tasks can be handled in one framework.	Training is data-hungry and task balancing is difficult.
MLLM-based reasoning segmentation	LISA, GLaMM (Lai et al., 2024; Rasheed et al., 2024)	Complex natural-language instructions, commonsense reasoning, and pixel grounding become possible.	Hallucination, fine boundary accuracy, heavy training cost, and evaluation ambiguity remain major problems.
Video, 3D, and complex-scene extensions	To be added	Closer to real applications with temporal or spatial consistency.	Annotation cost, memory, computation, and consistency evaluation are harder than in single-image settings.

category-level semantics (Liang et al., 2023; Xu et al., 2023b; Kirillov et al., 2023).

5.2.2. Training-based adaptation versus training-free composition

Compare OVSeg, SAN, and ESC-Net with PnP-OVSS and Trident. The former can learn task-specific structure, while the latter reduces training cost but may depend on heuristic fusion and multiple off-the-shelf models (Liang et al., 2023; Xu et al., 2023b; Lee et al., 2025; Luo et al., 2024; Shi et al., 2025).

5.2.3. Task-specific segmentation versus generalist reasoning

Compare specialized OVSS models with X-Decoder, LISA, and GLaMM. The main point is that generality improves interaction and reasoning ability but complicates training, evaluation, and deployment (Zou et al., 2023; Lai et al., 2024; Rasheed et al., 2024).

Figure plan: plot methods on axes such as semantic openness, mask precision, reasoning ability, and computational cost.

Figure 10: Planned figure: capability trade-off map.

6. Challenges and Future Directions

6.1. From open vocabulary to reliable open-world segmentation

Writing plan: argue that recognizing unseen category names is not enough. Future systems must handle label ambiguity, synonyms, rare categories, domain shift, and negative prompts.

6.2. Integrating semantic alignment and mask priors

Writing plan: discuss why CLIP, SAM, DINO, diffusion, and MLLM components are often complementary. Use ESC-Net and Trident as examples of 2025 methods that explicitly combine foundation-model strengths (Lee et al., 2025; Shi et al., 2025).

6.3. Reducing hallucination in reasoning segmentation

Writing plan: discuss hallucination and faithfulness in LISA- and GLaMM-style systems (Lai et al., 2024; Rasheed et al., 2024). Separate language plausibility from pixel-level correctness.

6.4. Evaluation beyond mIoU

Writing plan: argue that mIoU is insufficient for reasoning segmentation, grounded conversation, video, and 3D grounding. Future evaluation should include language faithfulness, grounding faithfulness, temporal consistency, and efficiency.

6.5. Video, 3D, and multi-context grounding

Writing plan: describe temporal consistency, spatial consistency, and multi-context reasoning as natural extensions. Add citations for VRS-HQ, GLUS, MC-Bench, or related papers after those source files are incorporated into the local bibliography.

Figure plan: organize future challenges by semantic complexity and spatial-temporal complexity, then place open-vocabulary segmentation, reasoning segmentation, video, 3D, and multi-context grounding on the map.

Figure 11: Planned figure: open problems map.

7. Conclusion

Writing plan: close with a bounded synthesis. State that open-vocabulary and reasoning segmentation has moved from CLIP-based category transfer in 2023, through SAM and unified decoders, to MLLM-driven reasoning and pixel grounding in 2024–2025. Emphasize that current systems still face trade-offs among semantic openness, mask quality, reasoning reliability, computational cost, and extension to video or 3D scenes. The conclusion should not introduce new methods; it should summarize the taxonomy and the main open challenges.

References

He, K., Gkioxari, G., Dollár, P., Girshick, R., 2017. Mask r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV).

Kirillov, A., He, K., Girshick, R., Rother, C., Dollár, P., 2019. Panoptic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollar, P., Girshick, R., 2023. Segment anything, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).

Lai, X., Tian, Z., Chen, Y., Li, Y., Yuan, Y., Liu, S., Jia, J., 2024. Lisa: Reasoning segmentation via large language model, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Lee, M., Cho, S., Lee, J., Yang, S., Choi, H., Kim, I.J., Lee, S., 2025. Effective sam combination for open-vocabulary semantic segmentation, in:

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 26081–26090.

Liang, F., Wu, B., Dai, X., Li, K., Zhao, Y., Zhang, H., Zhang, P., Vajda, P., Marculescu, D., 2023. Open-vocabulary semantic segmentation with mask-adapted clip, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Long, J., Shelhamer, E., Darrell, T., 2015. Fully convolutional networks for semantic segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

Luo, J., Khandelwal, S., Sigal, L., Li, B., 2024. Emergent open-vocabulary semantic segmentation from off-the-shelf vision-language models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I., 2021. Learning transferable visual models from natural language supervision, in: Proceedings of the International Conference on Machine Learning (ICML).

Rasheed, H., Maaz, M., Shaji, S., Shaker, A., Khan, S., Cholakkal, H., Anwer, R.M., Xing, E., Yang, M.H., Khan, F.S., 2024. Glamm: Pixel grounding large multimodal model, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Shi, Y., Dong, M., Xu, C., 2025. Harnessing vision foundation models for high-performance, training-free open vocabulary segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 23487–23497.

Xu, J., Liu, S., Vahdat, A., Byeon, W., Wang, X., De Mello, S., 2023a. Open-vocabulary panoptic segmentation with text-to-image diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Xu, M., Zhang, Z., Wei, F., Hu, H., Bai, X., 2023b. Side adapter network for open-vocabulary semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

Zou, X., Dou, Z.Y., Yang, J., Gan, Z., Li, L., Li, C., Dai, X., Behl, H., Wang, J., Yuan, L., Peng, N., Wang, L., Lee, Y.J., Gao, J., 2023. Generalized decoding for pixel, image, and language, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).